Design and Additive Manufacturing of Multipermeability Magnetic Cores

Lanbing Liu, Chao Ding, Shengchang Lu, Ting Ge, Yi Yan, Yunhui Mei, Khai D. T. Ngo, and Guo-Quan Lu

***Abstract*—Uneven distribution of magnetic flux in a conventional magnetic core limits the improvement of power density of the magnetic component. By making a magnetic core comprised of materials with different permeabilities, henceforth called multipermeability core, a more uniform flux distribution can be achieved without complex configuration and geometry of the core and winding. In this paper, we designed a three-ring toroid core to produce a flux uniformity factor, α, defined as $B_{\min}/B_{\max}$, in each of the three rings by increasing permeability from the inner to outer ring. With the same inductance, volume of the three-permeability toroid core was smaller than that of a single-permeability toroid core. For ease of fabricating multipermeability magnetic cores, a commercial multiextruder paste-extrusion 3-D printer was explored to process different magnetic pastes into 3-D structures. The magnetic pastes were formulated in our laboratory. Two types of magnetic paste systems, a high-temperature (>900 °C) pressure-less sinterable ferrite and a low-temperature (<250 °C) pressure-less curable powdered-iron, were developed and tested in the 3-D printer. By varying the magnetic/organic composition in each of the two material systems, we produced pastes that are compatible with the 3-D printer and can be processed into core materials with relative permeability ranging from 20 to 70. With the same dimensions, a 3-D-printed three-permeability toroid core had higher inductance than a single-permeability core.**

***Index Terms*—Additive manufacturing, custom magnetic cores, high-power density, uniform flux distribution.**

Manuscript received November 16, 2017; revised January 17, 2018; accepted March 8, 2018. Date of publication March 25, 2018; date of current version July 17, 2018. Paper 2017-PEDCC-1428.R1, presented at the 2017 IEEE Energy Conversion Congress and Exposition, Cincinnati, OH, USA, Oct. 1–5, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Power Electronic Devices and Components Committee of the IEEE Industry Applications Society. This work was supported by the National Science Foundation under Grant 1507314. *(Corresponding author: Lanbing Liu.)*

L. Liu and Y. Yan are with the Texas Instruments Incorporated Santa Clara, CA, USA (e-mail: lanbing@vt.edu; yiy87@vt.edu).

C. Ding is with the , Department of Materials Science and Engineering Virginia Tech, Blacksburg, VA 24061 USA (e-mail: dingchao@vt.edu).

S. Lu is with , the Bradley Department of Electrical and Computer EngineeringVirginia Tech, Blacksburg, VA 24061 USA (e-mail: lsheng1@vt.edu).

T. Ge and K. D. T. Ngo are with the , Center for Power Electronics Systems, and the Bradley Department of Electrical and Computer EngineeringVirginia Tech, Blacksburg, VA 24061 USA (e-mail: gting@vt.edu; kdtn@vt.edu).

Y. Mei is with the , School of Materials Science and EngineeringTianjin University, Tianjin 300072, China (e-mail: yunhui@tju.edu.cn).

G.-Q. Lu is with the , Department of Materials Science and Engineering, the Center for Power Electronics Systems, and the Bradley Department of Electrical and Computer EngineeringVirginia Tech, Blacksburg, VA 24061 USA, and also with the , School of Materials Science and EngineeringTianjin University, Tianjin, 300072 China (e-mail: gqlu@vt.edu).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TIA.2018.2819963

I. Introduction

As MAGNETIC components typically take up a significant portion of size and weight of a power converter, innovations leading to compact designs of magnetic components can greatly improve power density of the system [1]. In conventional magnetic cores, a crucial factor that limits the magnetic density is the uneven magnetic flux distribution in the core [2]. For example, in a toroid core, magnetic flux density is inversely proportional to the radius, leading to loss of energy density from both reduced permeability of inner rings as well as incompletely utilized energy storage of outer rings [3]. Having uniform distribution of magnetic flux throughout an inductor core, the volume of the inductor can be reduced without sacrificing its inductance, leading to a higher power density and efficiency. A further benefit of core volume reduction is the lower winding loss from reduced winding length.

In order to reduce volume and increase power density of a magnetic component, several techniques have been reported [3]–[6] to improve the uniformity of magnetic flux distribution throughout the core. For example, the concept of constant-flux inductor (CFI) was demonstrated in [3]–[5] to achieve relatively uniform magnetic flux distribution in the magnetic core by unconventional designs of the core shape and winding. The CFI in [3] with the core enclosed by an intricate winding structure comprises of several concentric toroid cells having a winding threaded into different slots separating the cells; the other CFI introduced in [4] consists of a spiral winding embedded in a magnetic core; to generate the uniform flux density, the winding width and spacing have to be precisely varied for each turn of the spiral. These intricate configurations and geometries of cores and windings are difficult to fabricate, thus posing barriers for their implementation.

Another method for making CFIs without a complicated winding structure is by engineering magnetic properties of the core. As demonstrated in [6], an inductor with a simple winding structure can have a uniform flux distribution by changing the magnetic permeability in the core. The researchers in [6] used low-temperature cofired ceramics (LTCC) technology to sinter together a laminated stack of ferrite tapes that produced different permeabilities. The LTCC technology requires a sintering temperature greater than 800 °C [7]–[9] and a pressure-assisted lamination step prior to sintering to bond multiple layers of green tapes [10]–[12]. Control of the lamination pressure and temperature is important to the final dimensions, densities, and other

0093-9994 © 2018 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.

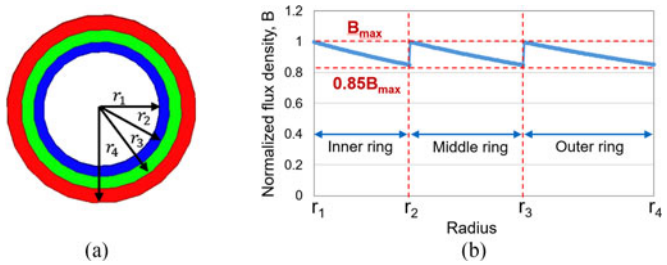


Fig. 1. Design of a three-ring toroid core by using different permeability materials to obtain more uniform distribution of flux density in the core. (a) Top view of the core structure and (b) normalized flux density along the radius direction.

properties of the sintered parts [13]–[16]. We view the LTCC technology to be a novel but limited approach for shaping the flux distribution in a magnetic core. The limitation comes from difficulties of finding green tapes with a wide range of magnetic properties, structural constraint from the 2-D tape, and requirement of pressure-assisted lamination.

Additive manufacturing (3-D-printing) has widely been demonstrated in various industries as a flexible and innovative manufacturing technology. However, only recently its application in power electronics is being explored [17]–[22]. Of the many additive manufacturing platforms we reviewed [22]; we narrowed down to a multiextruder 3-D printing platform that utilizes material feedstock in the form of paste [22], [23] for fabricating magnetic cores with varying permeabilities. Earlier, we had used the printing platform to make planar and toroid inductors by printing a magnetic paste for the core and a silver paste for the winding [20] and [21].

In this paper, we focused on proving the feasibility of printing multipermeability cores to improve flux distribution [24]. For the printer feedstock, we formulated magnetic pastes with varying compositions to give a wide range of permeabilities. Two different material systems were developed: 1) a high-temperature (>900 °C) pressure-less sinterable ferrite; and 2) a low-temperature (<250 °C) pressure-less curable powdered-iron. For each of the two material systems, at least three paste formulations were prepared. For feasibility demonstration, we fabricated three-permeability ferrite toroid cores and three-permeability powdered-iron toroid cores. To obtain a more uniform flux density in the toroid cores, the inner ring was printed using a paste with the lowest permeability while the outer ring with the highest permeability.

In the following, we first describe design considerations for making the three-permeability toroid inductor. Then, we introduce the multiextruder 3-D printer and describe details of the two types of magnetic pastes, procedure of printing the cores, and measured results of the cores.

II. Design of a Three-Permeability CFI

In this section, a toroid core with more uniform flux density, referred to by others as CFI is designed by using materials with three different permeabilities. The CFI core is to have the same inductance as that of a toroid core made of a single-permeability material. To illustrate the design procedure, we show in Fig. 1 a toroid-CFI core consisting of three concentric rings. The radius and permeability of each ring are determined as follows.

In a toroid core made of a given material, the flux density, B, decreases with radius according to

$$B = \frac{\mu_0 \mu_r n I}{2\pi r} \tag{1}$$

where μ_0 is the permeability of vacuum, μ_r is the relative permeability, n is the number of winding turns, I is the current flowing through the winding, and r is the radius from the core center. To make a toroid inductor from a single-permeability material, the ratio of its outer radius to inner radius is generally large to minimize the empty space of the center winding window and have enough core volume to meet the inductance requirement. The result is a highly nonuniform flux density throughout the core leading to an inefficient material use. However, in the three-ring CFI core, B is only nonuniform within each of the rings. From the inner ring to the outer ring, the permeability of each is progressively increased to compensate the drop in B with the increased radius as shown in Fig. 1(b). The permeability of each ring is chosen to give the same maximum flux density $B_{\max}$ and minimum flux density $B_{\min}$. The two B-values are calculated by

$$B_{\max} = \frac{\mu_0 \mu_{r_{\text{inner ring}}} n I}{2\pi r_1} = \frac{\mu_0 \mu_{r_{\text{middle ring}}} n I}{2\pi r_2} = \frac{\mu_0 \mu_{r_{\text{outer ring}}} n I}{2\pi r_3} \tag{2}$$

$$B_{\min} = \frac{\mu_0 \mu_{r_{\text{inner ring}}} n I}{2\pi r_2} = \frac{\mu_0 \mu_{r_{\text{middle ring}}} n I}{2\pi r_3} = \frac{\mu_0 \mu_{r_{\text{outer ring}}} n I}{2\pi r_4} \tag{3}$$

Dividing (2) by (3), we arrive at the radius ratios

$$r_1 : r_2 : r_3 : r_4 = 1 : \frac{1}{\alpha} : \frac{1}{\alpha^2} : \frac{1}{\alpha^3} \tag{4}$$

where α, termed flux uniformity factor, is equal to $\frac{B_{\min}}{B_{\max}}$.

The following example compares the size of a three-ring CFI toroid core with that of a single-permeability toroid core having the same inductance. For the three-ring CFI toroid core, α is set to be 0.85 and the relative permeabilities are, respectively, 25, 29.4, and 34.6, satisfying (2) and (3). The inner radius r_1 is set as 4 mm, then r_2, r_3, and r_4 (outer most radius), as shown in Fig. 1(a), are found from (4) to be 4.7, 5.5, and 6.5 mm, respectively. The single-permeability toroid core having the same inductance as the three-ring CFI core is designed to have a permeability of 25 and an inner radius of 4 mm. This design ensures that the maximum flux densities in both the cores are same. For both to have the same inductance, the outer radius of the single-permeability toroid core is found to be 8.3 mm, which is 28.2% larger than that of the CFI core. To illustrate the flux distributions in the two types of cores, we did 2-D simulations with the calculated geometry and permeability information. Fig. 2 shows the simulated results for the CFI core shown in (a) and the single-permeability core in (b).

Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.

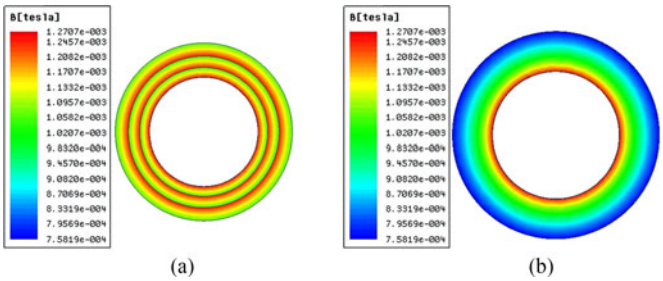


Fig. 2. 2-D simulated flux distributions using a set of geometry and permeability inputs derived for (a) a three-permeability toroid inductor with flux uniformity factor of 0.85, which has the same inductance as that of (b) a single-permeability toroid inductor. The latter is 28.2% larger and has a significantly nonuniform flux distribution.

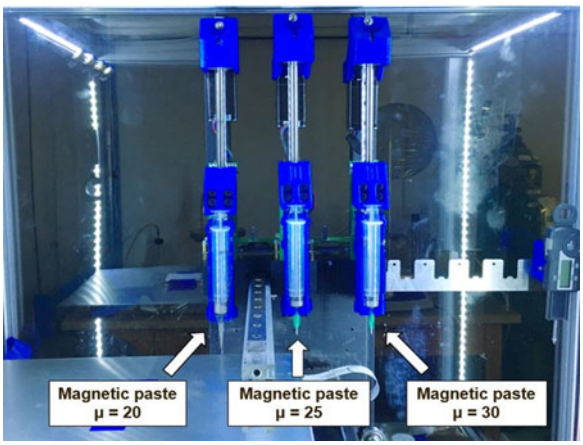


Fig. 3. Picture of a commercial multiextruder paste extrusion 3-D printer used for making multipermeability inductor cores.

III. Experiment

A. Commercial Multiextruder Paste Extrusion 3-D Printer

Shown in Fig. 3 is the multiextruder paste-extrusion 3-D printer (System 30 M from Hyrel 3-D) used to fabricate multipermeability magnetic cores. The printer is capable of coprocessing multiple paste materials through different extruders on a single printing platform.

B. Two Magnetic Paste Systems

1) High-Temperature (>900 °C) Sinterable Ferrite Paste: The composition of our ferrite paste consists of pentaerythritol tetraacrylate monomer (from Sigma-Aldrich Co. LLC) solution and Ni–Cu–Zn powder (from Powder Processing & Technology, LLC). The ferrite powder has an average particle size of 0.5 μm. Suitable volume fraction of the Ni–Cu–Zn powder was uniformly mixed in the monomer solution to obtain a paste that can be extruded out smoothly by the 3-D printer.

As shown in Fig. 4(a), a toroid-shaped structure was 3-D-printed from a ferrite paste. Ferrite pastes with different fractions of magnetic content were formulated and printed into toroid cores. They were then heated to around 920 °C under ambient pressure to burn out the organic compounds and densify or sinter. After sintering, the cores were characterized. Fig. 4(b) shows the plots of relative permeability versus frequency of the three

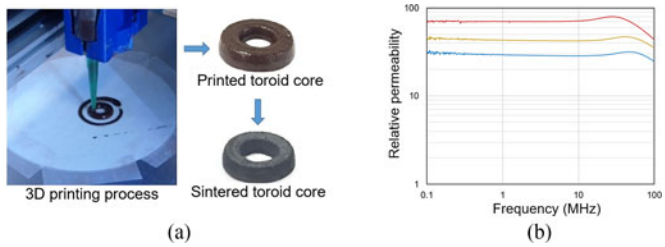


Fig. 4. (a) Demonstration of a 3-D-printed toroid core from a ferrite paste. (b) Plot of relative permeability versus frequency from several compositions of the ferrite paste system.

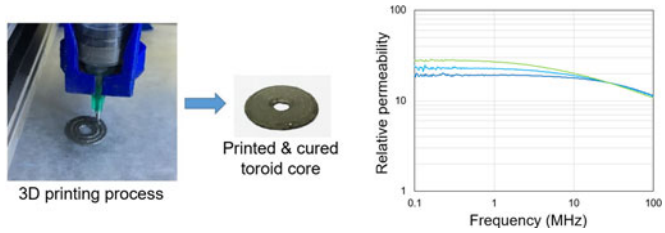


Fig. 5. (a) Demonstration of a 3-D-printed toroid core from a powdered-iron paste. (b) Plot of relative permeability versus frequency from several compositions of the powdered-iron paste system.

sintered cores. With the various paste compositions of the ferrite system, ferrite cores can be made to have relative permeability ranging from 28 to 70 at 1 MHz.

2) Low-Temperature Curable Powdered-Iron Paste: The composition of our powdered-iron paste consists of trimethylolpropane triacrylate (TMPTA) (from Sigma-Aldrich Co.LLC) monomer solution and two kinds of magnetic fillers. One of the magnetic fillers is a permalloy powder (from ESPI Metals) with an average particle size of 12 μm, and the other is flake-shaped Metglas 2705 M powder with an average particle size of 150 μm. The magnetic fillers were uniformly mixed in the TMPTA monomer solution.

Fig. 5(a) shows printing of a powdered-iron paste into a toroid. To characterize magnetic properties of various formulations of the powdered-iron material, toroid cores were made by pouring the pastes into a mold and heated to 200 °C for one hour under ambient pressure to polymerize the TMPTA monomer. After demolding, the cores were characterized in an impedance analyzer. Fig. 5(b) shows the plots of relative permeability versus frequency from three paste compositions showing that the relative permeability of the powdered-iron system can be adjusted from 21 to 35.

C. Feasibility Demonstration of 3-D-Printing Three-Permeability Toroid Cores

Tests of 3-D-printing multipermeability cores were done by making three-permeability cores with our ferrite and powdered-iron pastes. For the ferrite cores, the uniformity factor was set as 0.65, while 0.85 for the powdered-iron. The corresponding permeability of each ring of the two types of cores, satisfying (2) and (3), are listed in Table I. Then from (4), a relationship of different rings' radii can be calculated. Listed in Table II are the core lateral dimensions, i.e., r_1, r_2, r_3, and r_4. The

Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DESIGNED PERMEABILITY OF EACH RING OF THE THREE-PERMEABILITY TOROID CORES

Ring	Initial Permeability	
	Ferrite Core	*Powdered-iron core*
Inner	30	20
Middle	45	24
Outer	70	28

TABLE II
RADII OF RINGS OF THE THREE-PERMEABILITY TOROID CORES

	Ferrite Core	Powdered-iron core
r_1 (mm)	1.15	5.35
r_2 (mm)	1.77	6.29
r_3 (mm)	2.15	7.35
r_4 (mm)	4.25	8.69

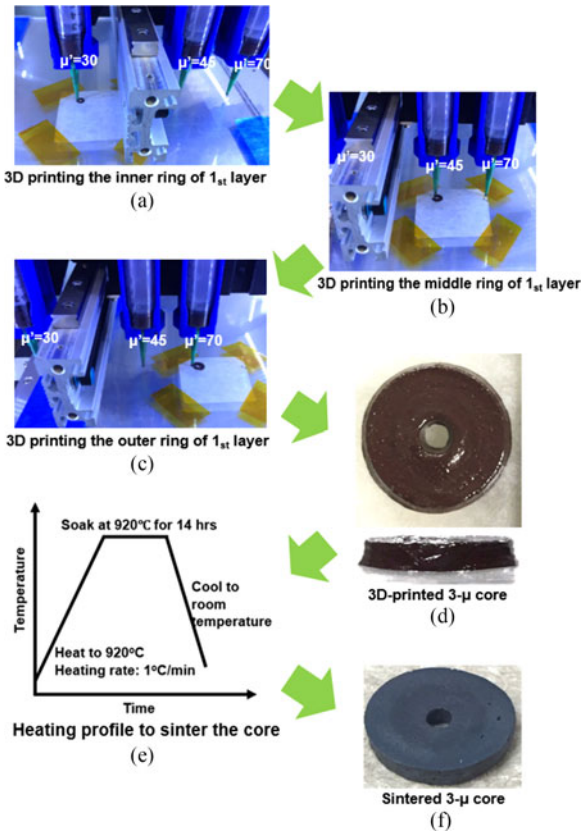


Fig. 6. Process of 3-D printing ferrite three-permeability toroid cores.

core heights are, respectively, 2 and 1.5 mm for the ferrite and powdered-iron.

Fig. 6 is a schematic diagram showing how a three-permeability ferrite toroid core was printed. Three ferrite pastes with permeabilities of 30, 45, and 70 were used to print the inner, middle, and outer rings, respectively. Single printing layer thickness was set to be 0.5 mm. Fig. 6(a)–(c) shows the process of building the first layer of a three-permeability ferrite toroid

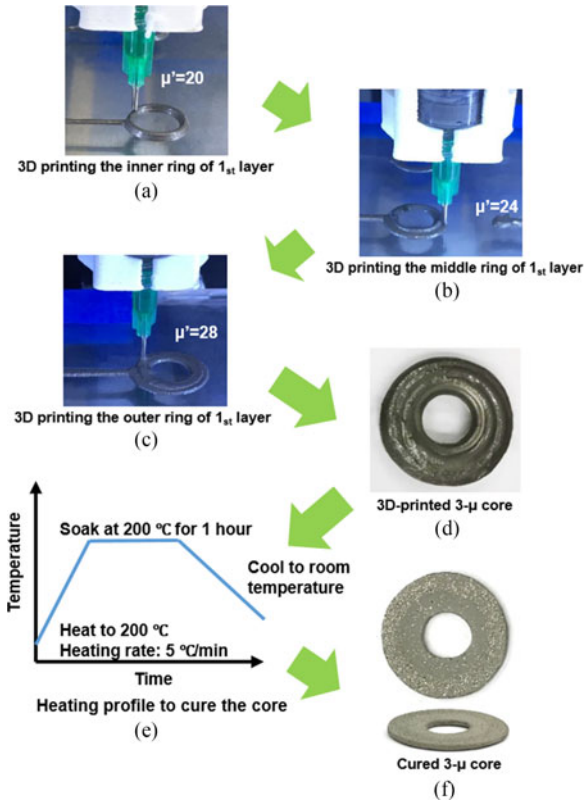


Fig. 7. Process of 3-D printing powdered-iron three-permeability toroid cores.

core. The process was then repeated to build the second to fourth layer until a thickness of 2 mm. Fig. 6(d) is a finished three-permeability ferrite toroid core. After the structure was built in the 3-D printer, it was heated under ambient pressure following a profile, as shown in Fig. 6(e) to burn out the organic compounds and sinter. The sintered three-permeability ferrite core is shown in Fig. 6(f). The ferrite core has a linear shrinkage of around 27% after sintering compared with the as-printed core.

The three-permeability powdered-iron cores were fabricated in a similar way, as shown in Fig. 7. After the structure was fabricated in the 3-D printer, it was cured at 200 °C for one hour under ambient pressure following the heating profile in Fig. 7(e). The cured toroid core, which has a linear shrinkage of around 5% after curing, is shown in Fig. 7(f).

For comparison, we also made two single-permeability toroid cores for each magnetic system: 1) one with the lowest permeability paste; and 2) the other with the highest. They have the same dimensions as their corresponding three-permeability cores and were made by the 3-D printer.

D. Inductance Charaterization

Cores were wire-wound into inductors with eight turns of windings for all ferrite inductors and four turns for all powdered-iron inductors. Plots of their inductance (L) versus frequency (f) over a frequency range from 1 kHz to 100 MHz at zero dc bias were obtained using a precision impedance analyzer (4294A; Agilent, Santa Clara, CA). Plots of their inductance (L) versus

Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.

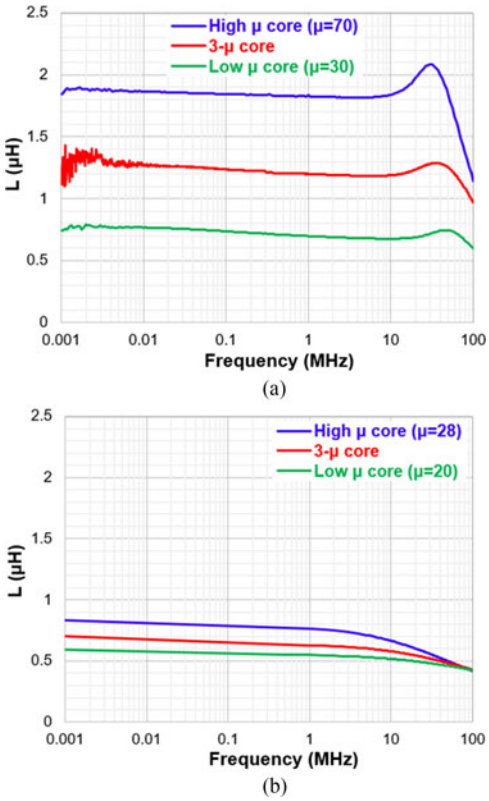


Fig. 8. Plots of inductance versus frequency measured on (a) ferrite and (b) powdered-iron inductors.

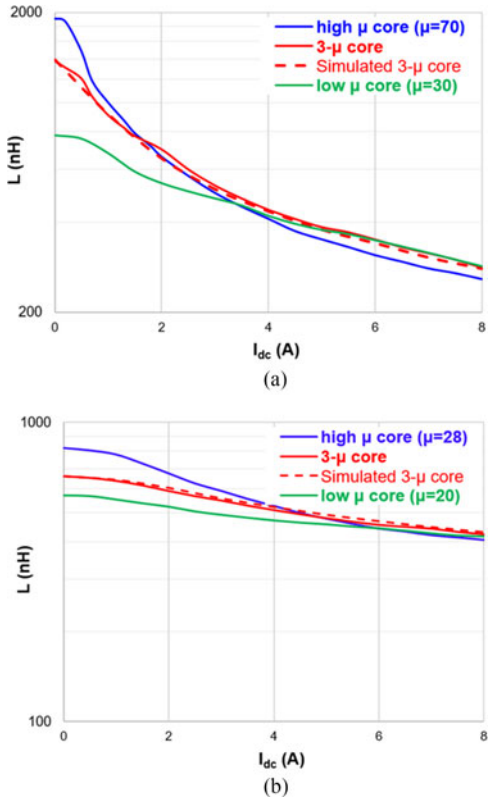


Fig. 9. Plots of inductance versus dc current measured on (a) ferrite and (b) powdered-iron inductors.

dc current (I_{dc}) were obtained by measuring L at 10 kHz using the same impedance analyzer with I_{dc} varied from 0 to 8 A.

E. Core-Loss Density Charaterization

The core-loss densities of the multipermeability and single-permeability cores were measured at room temperature as a function of B_{peak} at 1 MHz. A detailed description of the measurement principle and setup are given in [25].

IV. Results and Discussion

A. Inductance Versus Frequency

Shown in Fig. 8 are plots of inductance versus frequency of the three-permeability inductors and single-permeability inductors at 0 dc bias. The ferrite inductors had stable ($<$10% change) inductances until 10 MHz, and the resonance frequencies of all were $>$30 MHz. The powdered-iron inductors had stable ($<$10% change) inductances up to 1 MHz.

B. Inductance Versus DC Current

Fig. 9(a) shows the L-I_{dc} plots of the ferrite inductors. The initial inductance of the three-permeability inductor was lower than that of the single-permeability of 70 inductor due to its lower average initial permeability, but higher than that of the single-permeability of 30 inductor. Since all the cores had the same dimensions, the inductance density of the three-permeability inductor was around 70% higher than that of the 30-permeability inductor and 36% lower than that of the 70-permeability inductor at light load ($I_{dc} = 0$ A). With increasing dc current, the inductance of the three-permeability core decreased at a slower rate than that of the 70-permeability inductor and became higher at dc current greater than 1.7 A. The inductance density of the three-permeability inductor was around 11% higher than that of the 70-permeability inductor at heavy load ($I_{dc} = 8$ A). The inductance of the three-permeability inductor was always greater than that of the 30-permeability inductor over the entire range of dc bias.

The L-I_{dc} plots of the powdered-iron inductors are shown in Fig. 9(b). We see a similar behavior of the inductance versus dc bias exhibited by the powdered-iron as those by the ferrite described in the previous paragraph. The inductance density of the three-permeability inductor was around 16% higher than that of the 20-permeability inductor and 20% lower than that of the 28-permebility inductor at light load. At heavy load, inductance density of the three-permeability inductor was around 7% higher than that of the 28-permeability.

Also plotted in Fig. 9 are finite element analysis (FEA)-simulated L-I_{dc} curves of the three-permeability inductors. The input B–H loops of the materials used to make each ring were obtained by calculating from the measured L-I_{dc} curve of a single-permeability wire-wound toroid core made from the same material. As shown in these figures, for both ferrite and powdered-iron three-permeability inductors, the measured results are within 5% of the simulated.

Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.

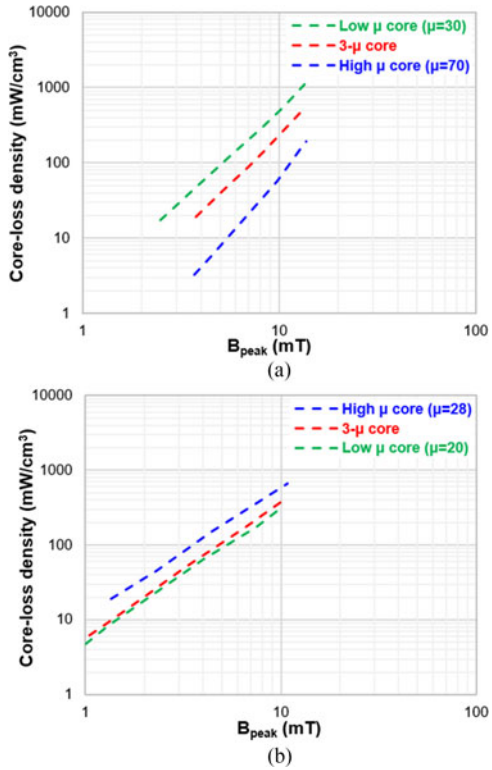


Fig. 10. Plots of core-loss density versus flux density on (a) ferrite and (b) powdered-iron cores measured at 1 MHz and 25 °C.

C. Core-Loss Density

Core-loss densities were measured and compared for the three-permeability and single-permeability inductors having the same dimensions. Fig. 10 shows plots of core-loss density versus B_{peak} of the ferrite and powdered-iron cores, both at 1 MHz. Of the two types of cores, the core-loss densities of the three-permeability cores were in between those of the high-μ and low-μ single-permeability cores. The core-loss density of the ferrite three-permeability core was around four times of the 70-permeabilty ferrite core and 40% of the 30-permeability ferrite core at 10 mT, while the core-loss density of the powdered-iron three-permeability core was around 1.2 times of the corresponding 20-permeabilty and 60% of the 28-permeability core at 10 mT.

V. Summary

We presented an additive manufacturing method for fabricating multipermeability magnetic cores to achieve a more uniform magnetic flux distribution. The printing platform is based on a commercial multiextruder 3-D printer to process magnetic pastes with different compositions into 3-D structures. Two paste systems were developed, a high-temperature (>900 °C) pressure-less sinterable ferrite and a low-temperature (<250 °C) pressure-less curable powdered-iron. Both were shown to be compatible with the 3-D printer and can be processed into cores with a wide range of relative permeabilities. For each magnetic material system, a three-ring toroid core was designed and 3-D printed. Characterization of inductance versus dc current showed that multipermeability inductors can increase inductance density by 7%–11% under heavy load over inductors made with a single-permeability material.

References

[1] C. R. Sullivan, D. Yao, G. Gamache, A. Latham, and J. Qiu, "(Invited) passive component technologies for advanced power conversion enabled by wide-band-gap power devices," *ECS Trans.*, vol. 41, no. 8, pp. 315–330, 2011.

[2] E. Snelling, *Soft Ferrites, Properties, and applications*. London, U.K.: Butter worth and Co Ltd, 1988.

[3] J. Seshadri, D. Xu, and K. Ngo, "Inductor structure with improved space utilization," *Electron. Lett.*, vol. 48, no. 3, pp. 164–165, 2012.

[4] H. Cui, K. D. Ngo, J. Moss, M. H. F. Lim, and E. Rey, "Inductor geometry with improved energy density," *IEEE Trans. Power Electron.*, vol. 29, no. 10, pp. 5446–5453, Oct. 2014.

[5] H. Cui, K. D. Ngo, J. Moss, M. Lim, and E. Rey, "Design and evaluation of the constant-flux inductor with enclosed-winding," in *Proc. 2014 IEEE Energy Convers. Congr. Expo.*, 2014, pp. 5229–5236.

[6] L. Wang, Z. Hu, Y.-F. Liu, Y. Pei, and X. Yang, "Multipermeability inductors for increasing the inductance and improving the efficiency of high-frequency DC/DC converters," *IEEE Trans. Power Electron.*, vol. 28, no. 9, pp. 4402–4413, Sep. 2013.

[7] M. Mu, W. Zhang, F. C. Lee, and Y. Su, "Laminated low temperature co-fired ceramic ferrite materials and the applications for high current POL converters," in *Proc. 2013 IEEE Energy Convers. Congr. Expo.*, 2013, pp. 621–627.

[8] R. Hahn, S. Krumbholz, and H. Reichl, "Low profile power inductors based on ferromagnetic LTCC technology," in *Proc. 56th Electron. Compon. Technol. Conf.*, 2006, pp. 528–533.

[9] K. Moon, S. Hong, H. Kim, and J. Kim, "A fabrication of dc–dc converter using LTCC NiZnCu ferrite thick films," in *Proc. Dig. IEEE Int. Magn. Conf.*, 2005, pp. 1109–1110.

[10] M. Mu, Y. Su, Q. Li, and F. C. Lee, "Magnetic characterization of low temperature co-fired ceramic (LTCC) ferrite materials for high frequency power converters," in *Proc. 2011 IEEE Energy Convers. Congr. Expo.*, 2011, pp. 2133–2138.

[11] S. M. Shapee, R. Alias, I. Azmi, Z. Ambak, M. Z. M. Yusoff, and M. R. Saad, "Microstructure for printed multilayer LTCC tape of different paste rheology," in *Proc. Mater. Sci. Forum*, 2010, vol. 654, pp. 2378–2381.

[12] A. ROOSEN, "Cold low pressure lamination of LTCC's," in *Proc. SPIE Proc. Ser.*, 2002, pp. 45–50.

[13] J. Macioszczyk, K. Malecha, A. Stafiniak, and L. J. Golonka, "Impact of processing parameters on the LTCC channels geometry," *Mater. Sci.-Poland*, vol. 33, no. 4, pp. 816–825, 2015.

[14] L. Li and Z. Wu, "Effect of lamination parameters on deformation energy of LTCC substrate based on finite element analysis," in *Proc. 2015 Int. Conf. Intell. Syst. Res. Mechatron. Eng.*, vol. 121, 2015, pp. 1569–1572.

[15] M. Franz *et al.*, "Material characteristics of the LTCC base material CeramTape GC," in *Proc. 2012 35th Int. Spring Semin. Electron. Technol.*, 2012, pp. 276–281.

[16] Z. Zhong, P. Arulvanan, and C. Goh, "Effects of key process conditions on warpage and via protrusion of LTCC substrate," *Mater. Manuf. Processes*, vol. 23, no. 2, pp. 182–187, 2008.

[17] M. Chinthavali, C. Ayers, S. Campbell, R. Wiles, and B. Ozpineci, "A 10-kW SiC inverter with a novel printed metal power module with integrated cooling using additive manufacturing," in *Proc. 2014 IEEE Workshop Wide Bandgap Power Dev. Appl.*, 2014, pp. 48–54.

[18] W. Liang, L. Raymond, and J. Rivas, "3-D-printed air-core inductors for high-frequency power converters," *IEEE Trans. Power Electron.*, vol. 31, no. 1, pp. 52–64, 2016.

[19] B. J. Kang, C. K. Lee, and J. H. Oh, "All-inkjet-printed electrical components and circuit fabrication on a plastic substrate," *Microelectron. Eng.*, vol. 97, pp. 251–254, 2012.

[20] Y. Yan, C. Ding, K. D. Ngo, Y. Mei, and G.-Q. Lu, "Additive manufacturing of planar inductor for power electronics applications," in *Proc. 3-D Power Electron. Integr. Manuf., Int. Symp.*, 2016, pp. 1–16.

[21] Y. Yan, J. Moss, K. D. Ngo, Y. Mei, and G.-Q. Lu, "Additive manufacturing of toroid inductor for power electronics applications," in *Proc. 2016 IEEE Energy Convers. Congr. Expo.*, 2016, pp. 1–6.

[22] Y. Yan, "Design methodology and materials for additive manufacturing of magnetic components," Ph.D. dissertation, Material Sci. Eng., Virginia Tech, Blacksburg, VA, USA, 2017.

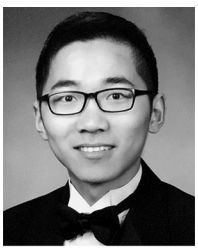
Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.

[23] J. A. Lewis, "Direct-write assembly of ceramics from colloidal inks," *Current Opinion Solid State Mater. Sci.*, vol. 6, no. 3, pp. 245–250, 2002.
[24] L. Liu *et al.*, "Design and additive manufacturing of multipermeability magnetic cores," in *Proc. 2017 IEEE Energy Convers. Congr. Expo.*, 2017, pp. 881–886.
[25] M. Mu, Q. Li, D. J. Gilham, F. C. Lee, and K. D. Ngo, "New core loss measurement method for high-frequency magnetic materials," *IEEE Trans. Power Electron.*, vol. 29, no. 8, pp. 4374–4381, Aug. 2014.



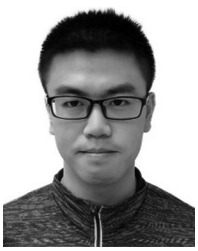
Lanbing Liu received the B.S. degree in material science and engineering from Tianjin University, Tianjin, China, in 2015. She is currently working toward the Ph.D. degree in materials science with Virginia Tech, Blacksburg, VA, USA.

Her research interest focuses on synthesis and processing high-frequency soft magnetic components for power electronics integration.



Chao Ding received the B.S. degree in materials science and engineering from Tianjin University, Tianjin, China, in 2016. He is currently working toward the Ph.D. degree in materials science and engineering with Virginia Tech, Blacksburg, VA, USA.

He is working on developing novel soft magnetic composites for power electronics application and improving manufacturing capability of magnetic components with 3-D printing technology.



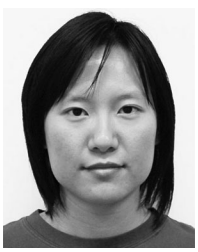
Shengchang Lu received the B.S. degree in energy and power engineering from the Huazhong University of Science and Technology, Hubei, China, in 2016. He is currently working toward the Ph.D. degree with the Center of Power Electronics Systems, Virginia Tech, Blacksburg, VA, USA.

His research interests include high-density magnetic components and high-frequency magnetic design.



Ting Ge (S'13) received the B.S. degree in electrical engineering from the Huazhong University of Science and Technology, Wuhan, Hubei, China, in 2012. He is currently working toward the Ph.D. degree in electrical engineering with the Center for Power Electronics Systems, Virginia Tech, Blacksburg, VA, USA.

His research interests include high-density magnetic components and high-frequency converter design.



Yi Yan received the B.S. degree in polymer materials science and engineering from the Wuhan Institute of Technology, Hubei, China, in 2009, the M.S. degree in material chemical engineering from Tianjin University, Tianjin, China, in 2012, and the Ph.D. degree in materials science and engineering from Virginia Tech, Blacksburg, VA, USA, in 2017.

She is currently with Texas Instruments Incorporated, Santa Clara, CA, USA, as a Packaging Engineer. She is working on the development of 3-D printing technology for power electronics applications, and magnetic components integration and package processes.



Yunhui Mei (M'12) received the B.S. and Ph.D. degrees in process equipment and controlling engineering from Tianjin University, Tianjin, China, in 2006 and 2010, respectively.

He is currently an Associate Professor with the Tianjin Key Laboratory of Advanced Joining Technology and the School of Material Science and Engineering, Tianjin University. He has authored or co-authored more than 60 papers on power electronic packaging. His current research interests include high-temperature packaging and materials for high-power-density applications.



Khai D. T. Ngo (S'82–M'84–SM'02–F'15) received the B.S. degree from California State Polytechnic University, Pomona, CA, USA, in 1979, and the M.S. and Ph.D. degrees from the California Institute of Technology, Pasadena, CA, USA, in 1980 and 1984, respectively, all in electrical and electronics engineering.

He is currently a Professor with the Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA, USA. He was a Member of Technical Staff at the General Electric Corporate Research and Development Center, Schenectady, NY, USA, from 1984 to 1988. Between 1988 and 2006, he was at the University of Florida, Gainesville, FL, USA. He is currently involved with technologies for integration and packaging of power-passive and active components to realize building blocks for power electronic systems at Virginia Tech. These technologies lead to power conversion systems with higher efficiency and higher power density. He also focuses on topologies, control, emission, and integration issues for radio frequency power converters. His other research interests include magnetic materials and components, energy reclamation, and power integrated circuits.



Guo-Quan (GQ) Lu received a double-major B.S. degree from Carnegie-Mellon University, Pittsburgh, PA, USA, in 1984, and the Ph.D. degree from Harvard University, Cambridge, MA, USA, in 1990, both in physics and material science and engineering.

He was at the Alcoa Technical Center before joining Virginia Tech, Blacksburg, VA, USA. Since 1998, he has worked extensively on research projects with the Center for Power Electronics Systems (CPES), Virginia Tech. He has authored or co-authored more than 100 peer-reviewed journal articles and more than 100 papers in conference proceedings. His research interests include synthesis and processing of functional materials, and development of electronic packaging technologies for power electronics integration.

Dr. Lu was the recipient of the Virginia Tech Sporn Award for excellence in teaching of engineering subjects and the National Science Foundation CAREER Award in 1995. His development of nanoscale silver paste, nanoTach, for low-temperature joining of semiconductor chips was recognized with a R&D 100 Award in 2007.

Authorized licensed use limited to: Danmarks Tekniske Informationscenter. Downloaded on October 07,2026 at 07:53:08 UTC from IEEE Xplore. Restrictions apply.